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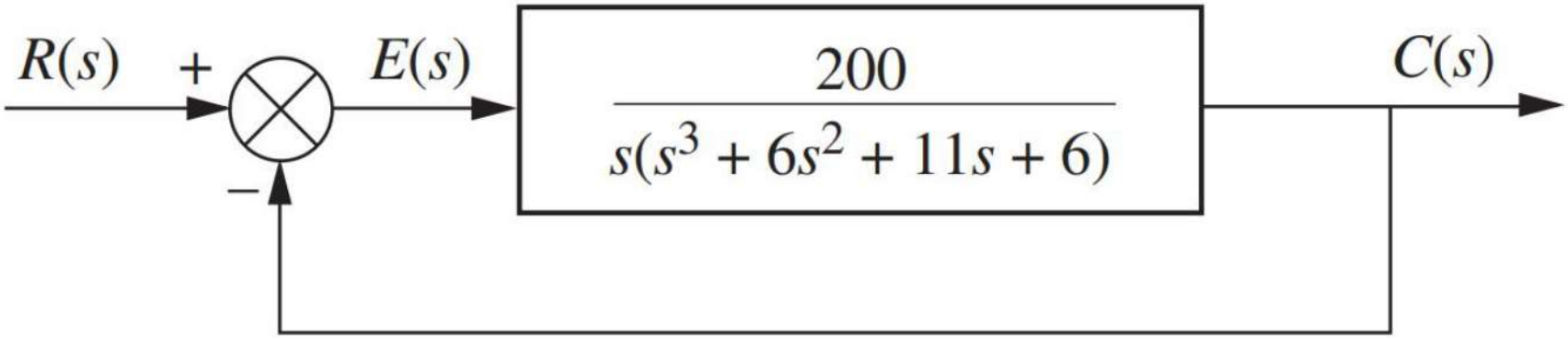
CW38

MAS508-G
Lab 5 – System stability



PROBLEM 1

Task: Given the following block diagram for the system shown below:



a) Find the closed-loop transfer function of the system.
b) Find the number of zeros and poles of the system.
c) Show the location of the poles in the root locus plot and conclude on the stability of the system.
d) Plot the step response of the system and see if it is according to your findings.

PROBLEM 2

Task: For a unity feedback system with the forward transfer function:

$$G(s) = \frac{K(s+20)}{s(s+2)(s+3)} \tag{2}$$

a) Find the range of K to make the system stable using the root locus.

PROBLEM 3

Task: For the following state-space system:

$$\dot{x} = \begin{bmatrix} 0 & 1 & 3 \\ 2 & 2 & 4 \\ 1 & -4 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u \tag{3}$$

$$y = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} x$$

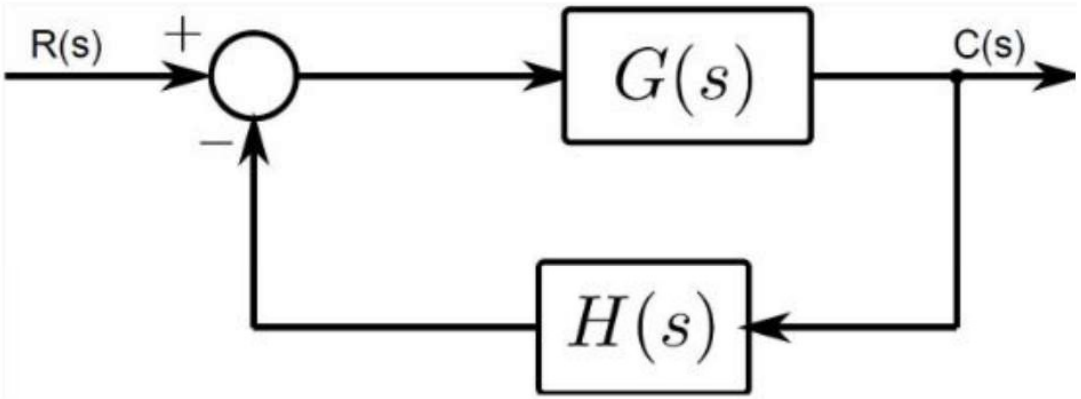
a) Find out eigenvalues of the system.

b) Identifying how many poles are in the left half-plane, in the right half-plane, and on the $j\omega$-axis. Conclude on system's stability.

PROBLEM 4

Task: Consider a unity feedback system ($H(s) = 1$) where

$$G(s) = \frac{K}{(s+1)(s+3)(s+4)} \tag{6}$$



a) Show an example of step response for the system when he is stable and unstable.
b) Determine the range of K for which the system is stable.
c) Show the frequency response of the system using the Bode diagram.
d) Check stability of the system using the Bode diagram by measure the gain margin and the phase margin.

PROBLEM 5

Task: Assuming the following unity feedback system:

$$G(s) = \frac{K}{(s^2 + 2s + 4)(s + 2)} \tag{7}$$

a) Draw the frequency response of the system using the Bode diagram.
b) Determine the gain and phase margins of the open-loop.
c) Find the range of gain $K > 0$ for stability and instability.